Linear Algebra for High Dimensional Water Data

Ping Wang

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Table of Symbols

Symbol	**Typical meaning**
a, b, c, x_1	Scalars are lowercase
$\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}, \mathbf{b}, \mathbf{a}_1$	Vectors are bold lowercase
$\mathbf{A}, \mathbf{B}$	Matrices are bold uppercase
$\mathbf{x}^T$, $\mathbf{A}^T$	Transpose of a vector or matrix
$\mathbf{A}^{-1}$	Inverse of a matrix
$\mathbf{x}^T\mathbf{y}$	Dot product of $\mathbf{x}$ and $\mathbf{y}$
$[\mathbf{b}_1\ \mathbf{b}_2\ \mathbf{b}_3]$	Matrix of column vectors stacked horizontally
$\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$	Set of vectors
$\mathbf{R}$	Real numbers
$\mathbb{R}^n$	n-dimensional vector space of real numbers
$\iff$	If and only if
$\implies$	Implies
$\mathbf{v} \in \mathbb{R}^n$	$\mathbf{v}$ is an element of set $\mathbb{R}^n$
$\mathbf{I}_m$	Identity matrix of size $m \times m$
$\mathbf{0}_{m,n}$	Matrix of zeros of size $m \times n$
$\mathbf{1}_{m,n}$	Matrix of ones of size $m \times n$
$\mathbf{e}_i$	Standard vector
span$\{\mathbf{b}_1\}$	Span (genearting set) of $\mathbf{b}_1$
det($\mathbf{A}$)	Determinant of $\mathbf{A}$
λ	Eigenvalue

1

Vectors and Vector Operations

1.1 What is a Vector?

Linear Algebra is the branch of math that primarily focuses on vectors, vector spaces, and linear transformations within vector spaces, which allows the manipulation of vectors through operations, such as addition and scalar multiplication. AI, machine learning, and robotics rely heavily on Linear Algebra.

A vector is a mathematical object. In contrast to a scalar (number), which carries only a magnitude, a vector represents a quantity that has both magnitude and direction. Vectors can be added together and multiplied by a scalar, and remain the same objects of the same type. Keep in mind that vectors are not just algebraic objects; they are also geometric objects.

1.1.1 Vectors as Ordered Lists of Numbers

Let's look at some of the Mg, K, and Ca data from the dataset that we've got from our ICP-MS lab for drinking water samples.

sample	**Mg**	**K**	**Ca**
TN1	3815	2265	2829
TN2	6909	1713	4168
TN3	3566	1411	2683
TN4	10083	1710	4899
TN5	3328	967	2523
CA1	10151	3428	3235
CA2	548	552	166
CA3	11166	3361	3484
VA1	2842	2704	1715
VA2	2746	2678	1073
MA1	725	788	334
MA2	5779	20289	2637
MA3	736	765	340

Table 1.1: Mg, K, and Ca Data

Table 1.1 has columns and rows, and stores the Mg, K, and Ca data in a tabular form. We say that the Mg, K and Ca data are 2-dimensional arrays. Arrays are numbers (or scalars) that are organized into lists. Arrays are used to store multiple values in a single variable instead of declaring separate variables for each value. The original water dataset has a large number of features (45 features) in columns and 29 samples in rows. We call this type of data high dimensional data. We can organize all the Mg data into a vector, as in

$$\mathbf{v} = \begin{bmatrix} 3815 \\ 6909 \\ 3566 \\ 10083 \\ 3328 \\ 10151 \\ 548 \\ 11166 \\ 2842 \\ 2746 \\ 725 \\ 5779 \\ 736 \end{bmatrix} \in \mathbb{R}^{13}$$

Algebraically, a vector is an ordered list of numbers. Each of the numbers in a vector is called a component (or coordinate). Since the vector above $\mathbf{v}$ has 13 components, we say that $\mathbf{v}$ is a 13-dimensional vector. Consequently, if $\mathbf{v}$ is a 13-dimensional vector, we say that $\mathbf{v} \in \mathbb{R}^{13}$ ($\mathbf{v}$ is in $\mathbb{R}^{13}$). $\mathbb{R}^{13}$ is the 13-dimensional vector space of real numbers.

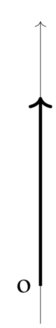
Vectors as numerical representations of quantitative attributes of data samples, can be used to train AI/ML models.

1.1.2 Vectors as Directed Line Segments

Geometrically, a vector is a directed line segment. A nonzero vector is a directed line segment whose starting point is at the origin. Any directed line segment specifies a vector, which is the parallel directed segment starting at the origin with the same direction and length.

A vector may belong to the real line $\mathbb{R}$, or to the plane $\mathbb{R}^2$, or to space $\mathbb{R}^3$.

Example 1.1 A vector in the line



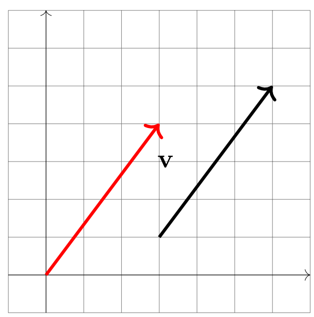
The 1-dimensional space $\mathbb{R}^1$ is a line, like the x axis or y axis.

Example 1.2 Vectors in $\mathbb{R}^2$

The Euclidean plane, usually denoted $\mathbb{R}^2$, is the set of ordered pairs (x,y) of real numbers.$\mathbb{R}^2$, is the entire xy plane. We say that $\mathbb{R}^2$ is a two-dimensional vector space. $\mathbf{v}$ is an element of $\mathbb{R}^2$. For instance,

$$\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \in \mathbb{R}^2$$

$\mathbf{v}$ is a 2-dimensional vector. Since $\mathbf{v}$ has two components, we say that $\mathbf{v} \in \mathbb{R}^2$ ($\mathbf{v}$ is in $\mathbb{R}^2$). Vectors in $\mathbb{R}^2$ are drawn as directed line segments going from the origin (0,0) to the corresponding point. The end point of $\mathbf{v}$ is (3,4). When reresenting vectors in $\mathbb{R}^n$ as directed line segments, they do not have to start at the origin. For instance, in $\mathbb{R}^2$, the vector $\mathbf{v}$ can be represented by any directed line segments as long as they have the same length and same direction. Both in the following plot below represent the vector $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$:

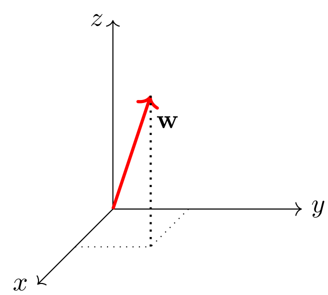


A vector can start anywhere regardless of the origin. The set of all possible 2-dimensional vectors (the whole plane) is the vector space $\mathbb{R}^2$, represented by the xy plane.

Example 1.3 Vectors in $\mathbb{R}^3$

In geometry, a three-dimensional coordinate system consists of three axes: the x-axis, the y-axis, and the z-axis. The three axes point in perpendicular directions to each other. An element of $\mathbb{R}^3$ is

$$\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \in \mathbb{R}^3.$$

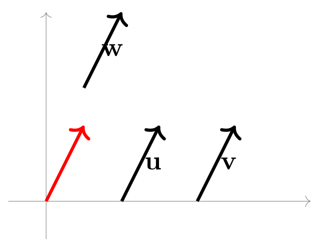


The set of all possible 3-dimensional vectors is vector space $\mathbb{R}^3$.

Example 1.4 Equivalent vectors

Two vectors are equal (or equivalent) if each individual component of one vector must match with the corresponding component of the other vector. Essentially, they must have the same the magnitude and direction in every axis.

As long as vectors that point in the same direction and have the same length they are equivalent vectors regardless of their staring point or position. Every vector is equivalent to some vector in standard position with its tail at the origin.



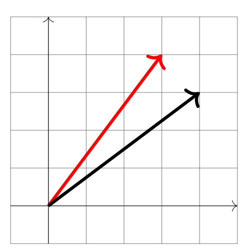
As in the plot above, vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are all equivalent. We write

$$\mathbf{u} = \mathbf{v} = \mathbf{w} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Example 1.5 The following two vectors are not equivalent.

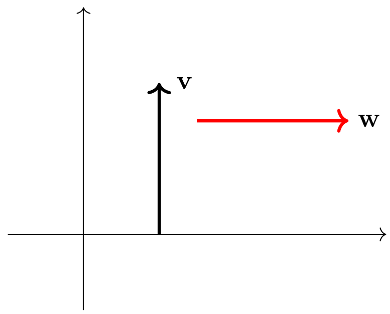
$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} \neq \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

The two vectors above are not equivalent because the corresponding components in two vectors are not the same, as in



Example 1.6 Transpose of a vector

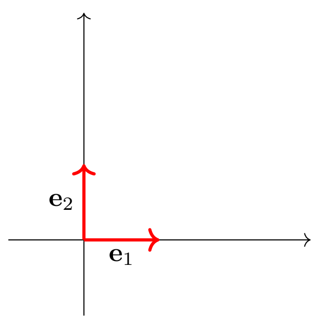
$\mathbf{v}$ is not equivalent to $\mathbf{w}$ as in the following plot. Actually, $\mathbf{v}$ is the transpose of $\mathbf{w}$, meaning that $\mathbf{w}$ is a row vector while $\mathbf{v}$ is a column vector. We write the transpose of $\mathbf{w}$ as $\mathbf{w}^T$. The transpose of a vector is with its rows and columns swapped, meaning it converts a column vector into a row vector, and vice verse; in simple terms, it flips the orientation of the vector while keeping the elements in the same order.



Example 1.7 Standard unit vectors

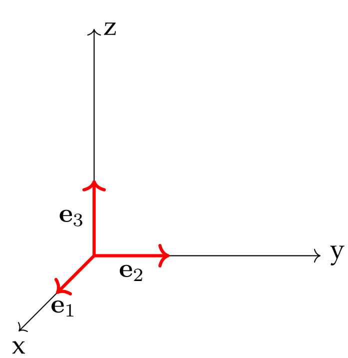
In $\mathbb{R}^2$, unit vectors in the positive directions of the coordinate axes are called the standard unit vectors. In $\mathbb{R}^2$, these vectors are denoted by

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \ \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



In $\mathbb{R}^3$, the standard unit vectors are denoted by

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \ \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \ \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



Example 1.8 The position of a target can be expressed as a vector.

$$\mathbf{p} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}.$$

The position vector $\mathbf{p}$ represents the target 2 meters in front of the camera, 0 meters to the side, and 3 meters above the camera.

1.2 Vector Operations

Numerically, vector operations are component-wise. When performing an operation on two or more vectors, the operation is applied indepdently to each corresponding component of the vectors, meaning each element in the same position within vectors is calculated seperately. Any component-wise operation on two or more vectors requires that the vectors have the same dimensionality.

Geometrically, two operations are defined on directed line segments: A directed line segments can be scaled (stretched or compressed), and two directed line segments with the same origin can be added using the parallelogram method. An addition of several scaled (stretched or compressed) vectors are called a linear combination of these vectors.

We can visualize 2-dimensional vectors and 3-dimensional vectors; however, it is hard to visualize high-dimensional vectors. Unless otherwise indicated, the formulas we use for 2-dimensional vectors also apply to n-dimensional vectors.

1.2.1 Vector Addition

Adding Corresponding Components

The components of $\mathbf{v} + \mathbf{u}$ can be found by adding the components of $\mathbf{v}$ and $\mathbf{u}$:

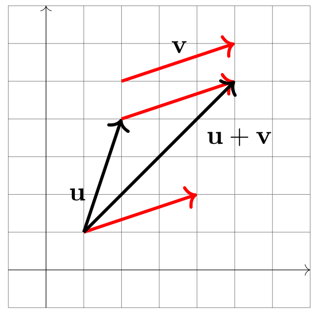
$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+3 \\ 2+1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

We can add vectors of higher dimensions, for example

$$\begin{bmatrix} -2 \\ 1 \\ 4 \\ 7 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \\ 0 \\ -2 \\ -3 \end{bmatrix} = \begin{bmatrix} -2+3 \\ 1-1 \\ 4+0 \\ 7-2 \\ 5-3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 4 \\ 5 \\ 2 \end{bmatrix}$$

Adding Vectors Represented as Directed Line Segments

To add vectors $\mathbf{a}$ and $\mathbf{b}$, you move (translate) $\mathbf{b}$ so that it starts at the end of $\mathbf{a}$. As you move $\mathbf{b}$, keep its length and direction the same. As long as you do not change an arrow's length or direction, it represents the same vector. The sum of $\mathbf{a} + \mathbf{b}$ is represented by the directed line segment that goes from the start of $\mathbf{a}$ to the end of $\mathbf{b}$.

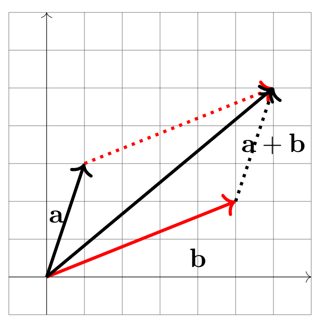


Numerically, the sum of two vectors is just the sum of the corresponding components:

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

Parallelogram Method

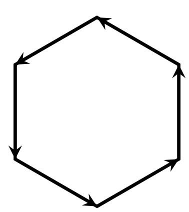
We can add two vectors by sliding vectors $\mathbf{a}$ and $\mathbf{b}$ so that their tails are at the same point and the two vectors determine a parallelogram. Applying the tip-to-tail addition method shows that the sum $\mathbf{a} + \mathbf{b}$ is the diagonal of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$.



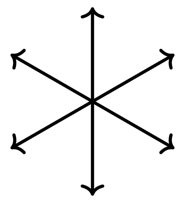
Numercially, the sum of $\mathbf{a}$ and $\mathbf{b}$ is

$$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

Example 1.9 The sum of six vectors gives $\mathbf{0}$.

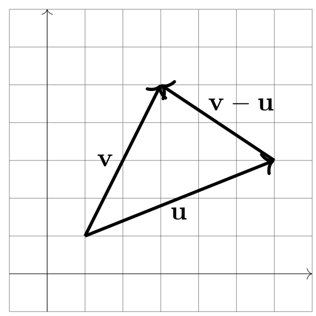


Example 1.10 The sum of following six vectors gives $\mathbf{0}$.



Example 1.11 Vector subtraction

$\mathbf{v} - \mathbf{u}$ is the same length but pointing in the opposite direction to $\mathbf{v}$, as in

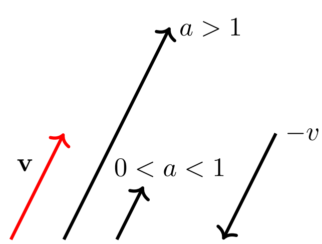


1.2.2 Scalar Multiplication

In the context of vectors, a number is called a scalar because it can be used to scale a vector via multiplication. Multiplying a vector $\mathbf{v}$ by a scalar is defined as multiplying each component of $\mathbf{v}$. For instance,

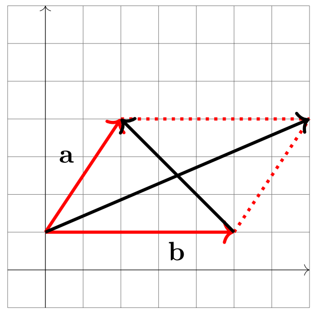
$$3 \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \\ 15 \end{bmatrix}$$

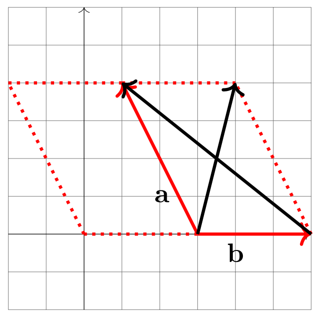
Multiplication of a vector by a scalar stretches or compresses the vector, and changes its direction in case that scalar is negative, as in



As can be seen from the plot above, mulpilication of a vector by a scalar a affects the length and the direction of the vector. The product $a\mathbf{v}$ stretches $\mathbf{v}$ when $a > 1$; and it compresses $\mathbf{v}$ when $0 < a < 1$. If $a < 0$, then it reverses the direction of $\mathbf{v}$ and it stretches when $a < -1$ and compresses when $-1 < a < 0$.

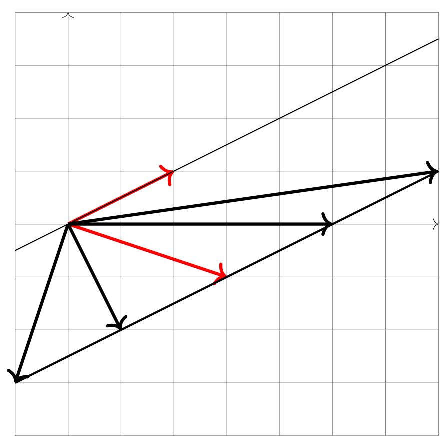
Example 1.12 Particular cases of tip-to-tail/parallelogram method





Example 1.13 The set of vectors $c\mathbf{v} + \mathbf{w}$ form a line.

$$\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \ \mathbf{w} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$



Properties of Vector Operations

The properties for vector addition and scalar multiplication are as follows:

1. $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$
2. $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$
3. $\mathbf{a} + 0 = \mathbf{a}$
4. $\mathbf{a} + (-\mathbf{a}) = 0$
5. $c(\mathbf{a} + \mathbf{b}) = c\mathbf{a} + c\mathbf{b}$
6. $(c + d)\mathbf{a} = c\mathbf{a} + d\mathbf{a}$
7. $(cd)\mathbf{a} = c(d\mathbf{a})$
8. $1\mathbf{a} = \mathbf{a}$

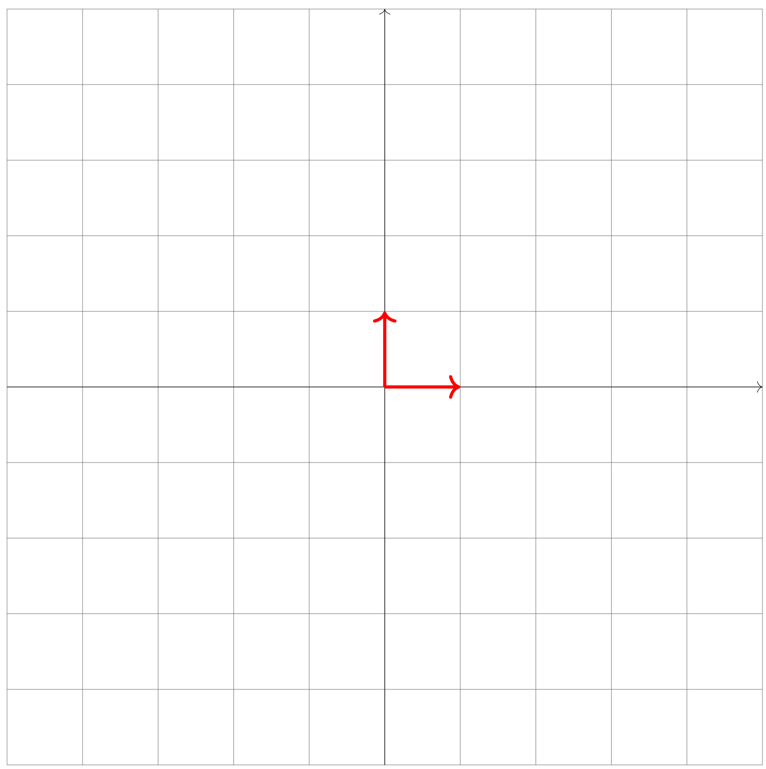
Example 1.14 Standard Bases for $\mathbb{R}^2$

We can use two standard orthogonal unit vectors: e_1, e_2, to construct an arbitrary plane.

$$\mathbf{e_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \ \mathbf{e_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\{\mathbf{e_1}, \mathbf{e_2}\}$ is the standdard basis for $\mathbb{R}^2$.

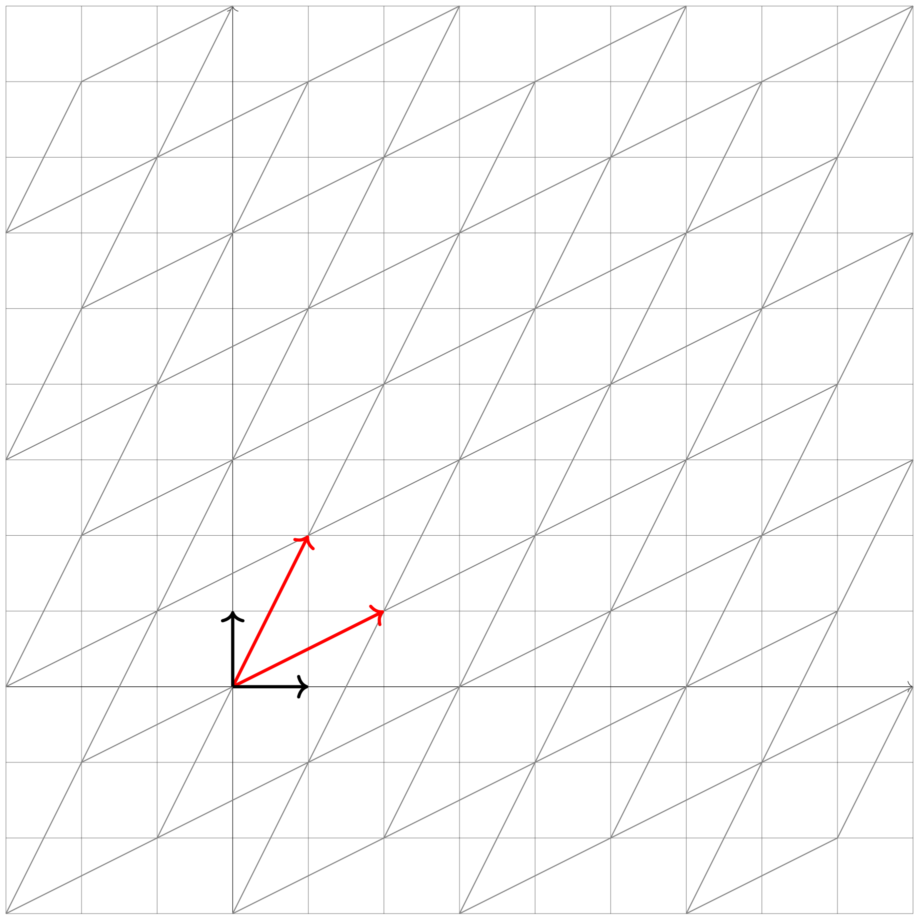
You can create any vector using a linear combination of these two standard basis vectors, where the weights are scalar values assigned to each basis vector. Essentially, you scale each basis vector by a number and add them together to get your desired vector.



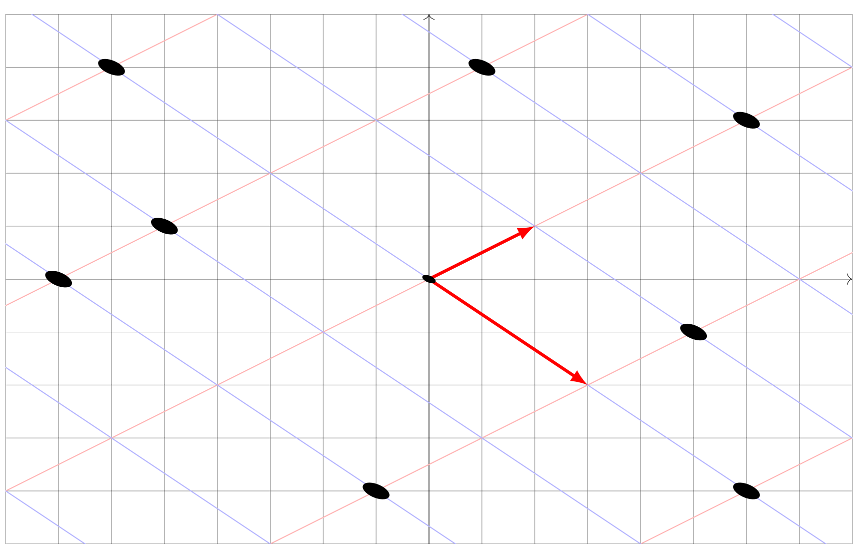
Example 1.15 Linear Combinations of a pair of vectors, $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ The linear combination of the vectors $\mathbf{v}$ and $\mathbf{w}$ with scalars c_1, c_2 is the vector

$$c_1\mathbf{v} + c_2\mathbf{w}$$

The scalars c_1, c_2 are called the weights of the linear combination. You can create any vector using a linear combination of these two basis vectors, $\mathbf{v}$ and $\mathbf{w}$.



Example 1.16 Express the labeled dots as linear combination of standard basis vectors and another basis vectors. Draw the lines paremetrically as $x_1\mathbf{v} + x_2\mathbf{w}$.



1.3 Dot Product

Dot product is an operation that takes two vectors and returns a scalar. The dot product of two vectors is a scalar value that represents the product of their magnitude multiplied by the cosine of the angle between them; essentially, it measures how much the two vectors are aligned in the same direction, with a positive result indicating alignment and a negative result indicating opposition. Sometimes the dot product is also called inner product.

Dot product can be used to measure the similarity between two vectors. If the dot product is zero, then the two vectors are orthogonal. Dot product can be used to project one vector onto another and compute the angle between two vectors, and calculating work done by a force in a specific direction.

1.3.1 Computing Dot Product in Terms of Components

To compute the dot product of two vectors is to multiply corresponding components and add the resulting products.

Example 1.17 Compute the dot product of two vectors.

$$\mathbf{u}^T\mathbf{v} = \mathbf{v}^T\mathbf{u} = \begin{bmatrix}1 & 2 & 3\end{bmatrix}\begin{bmatrix}3\\2\\1\end{bmatrix} = \begin{bmatrix}3 & 2 & 1\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix} = 10$$

Example 1.18 Compute the dot product of two n-dimensional vectors

Given $\mathbf{u} = \begin{bmatrix}u_1\\u_2\\\vdots\\u_n\end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix}v_1\\v_2\\\vdots\\v_n\end{bmatrix}$ are vectors in $\mathbb{R}^n$, the the dot product of the vectors $\mathbf{u}$ and $\mathbf{v}$ is defined as the scalar

$$\mathbf{u}^T\mathbf{v} = \mathbf{v}^T\mathbf{u} = u_1v_1 + u_2v_2 + ... + u_nv_n,$$

the sum of the products of corresponding components (entries).

Example 1.19 Sum of the elements of the vector

The dot product of a vector with the vector of ones gives the sum of the elements of the vector, as in

$$\mathbf{1}^T\mathbf{a} = a_1 + ... + a_n.$$

$$\begin{bmatrix}1 & 1 & 1 & 1 & 1\end{bmatrix}\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix} = 1+2+3+4+5 = 15$$

Example 1.20 Sum of squares

The dot product of a vector with itself gives the sum of squares of the elements of the vector, as in

$$\mathbf{a}^T\mathbf{a} = a_1^2 + ... + a_n^2.$$

$$\begin{bmatrix}1 & 2 & 3 & 4 & 5\end{bmatrix}\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix} = 1^2 + 2^2 + 3^3 + 4^2 + 5^2 = 55$$

Example 1.21 Selective sum

$\mathbf{b}^T\mathbf{a}$ is the sum of the elements in $\mathbf{a}$ for which $b_i = 1$ if all entries in $\mathbf{b}$ are either 0 or 1.

$$\begin{bmatrix}1 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix} = 1 + 5 = 6$$

1.3.2 Computing Dot Product by Using the Angle between Two Vectors

If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors in $\mathbb{R}^n$, and if θ is the angle between $\mathbf{u}$ and $\mathbf{v}$, then the dot product of $\mathbf{u}$ and $\mathbf{v}$ is denoted by

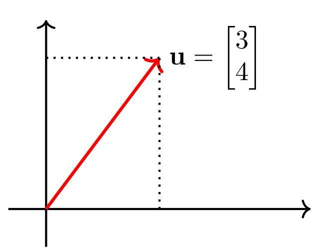
$$\mathbf{u}^T\mathbf{v} = \mathbf{v}^T\mathbf{u} = \|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta).$$

Length of a Vector

The length of a vector is the square root of the dot product of a vector with itself. The vector's length is also called the norm of the vector.

Example 1.22 Find the length of a vector.

$\left\|\begin{bmatrix}3\\4\end{bmatrix}\right\| = \sqrt{3^2 + 4^2} = 5$, as in



The length of a vector $\mathbf{u}$ in $\mathbb{R}^n$ is the number

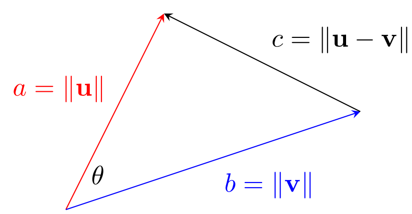
$$\|\mathbf{u}\| = \sqrt{\mathbf{u}^T\mathbf{u}} = \sqrt{u_1^2 + u_2^2 + ... + u_n^2}.$$

Angle Between Two Vectors

We define the angle between $\mathbf{u}$ and $\mathbf{v}$ to be the angle θ determined by $\mathbf{u}$ and $\mathbf{v}$ that satisfies the inequalities $0 \leq \theta \leq \pi$.

Example 1.23 Angle between two vectors

Geometrically, the dot product of $\mathbf{u}$ and $\mathbf{v}$ equals the length of $\mathbf{u}$ times of the length of $\mathbf{v}$ times the cosine of the angle between them: $\mathbf{u}^T\mathbf{v} = \|\mathbf{u}\| \cdot \|\mathbf{v}\| \cdot \cos(\theta)$.



Example 1.24 Orthogonal vectors

Two vectors are orthogonal to each other if their dot product is zero; that means that the projection of one vector onto the other collapses to a point. The two vectors are thus only the same if the two vectors have zero projection one on the other. Two vectors $\mathbf{u}$, $\mathbf{v}$ in $\mathbb{R}^n$ are orthogonal or perpendicular if $\mathbf{u}^T\mathbf{v} = 0$.

$\mathbf{u} \perp \mathbf{v}$ means $\mathbf{u}^T\mathbf{v} = \mathbf{v}^T\mathbf{u} = 0$.

Example 1.25 Determine if two vectors are orthogonal.

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \ \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\mathbf{u}^T\mathbf{v} = \mathbf{v}^T\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1 \cdot 0 + 0 \cdot 1 = 0.$$

Thus, the vectors are orthogonal because $\mathbf{u}^T\mathbf{v} = 0$.

Example 1.26 Compute the angle between two vectors.

Given $\mathbf{u} = \begin{bmatrix} 3 \\ 4 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$, compute the dot product of $\mathbf{u}$ and $\mathbf{v}$.

The dot product: $\mathbf{u}^T\mathbf{v} = 3 \times 1 + 4 \times (-2) + (-1) \times (-2) = -7$

The length of $\mathbf{u}$: $\|\mathbf{u}\| = \sqrt{3^2 + 4^2 + (-1)^2} = \sqrt{26}$

The length of $\mathbf{v}$: $\|\mathbf{v}\| = \sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{9} = 3$

$\cos(\theta) = \frac{\mathbf{u}^T\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} \approx -0.46$

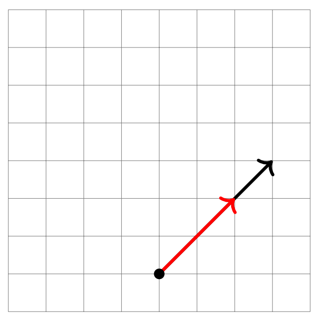
Thus, the angle between $\mathbf{u}$ and $\mathbf{v}$ is approximately $117°$.

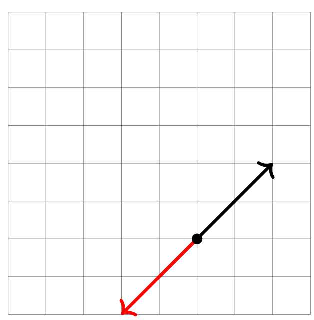
Example 1.27 Acute and obtuse angles

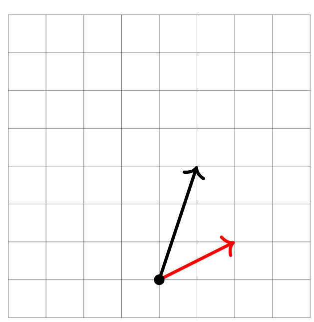
Suppose $\mathbf{a}$ and $\mathbf{b}$ are two nonzero vectors of the same size.

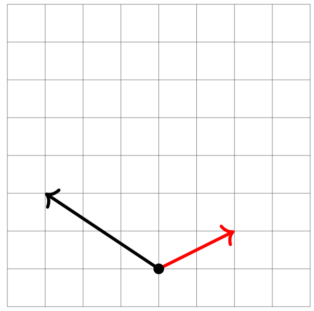
1. If the angle is $\pi/2 = 90\circ$, i.e., $\mathbf{a}^T\mathbf{b} = 0$. The vectors are said to be orthogonal. We write $\mathbf{a} \perp \mathbf{b}$. A zero vector is orthogonal to any vector.
2. If the angle is an acute angle, which means $\mathbf{a}^T\mathbf{b} > 0$, the vectors have positive dot product.
3. If the angle is an obtuse angle, which means $\mathbf{a}^T\mathbf{b} < 0$, the vectors have negative dot product.
4. If the angle is zero, then $\cos 0 = 1$, which means $\mathbf{a}^T\mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\|$. The vectors are said to be algined. Each vector is a positive multiple of the other.

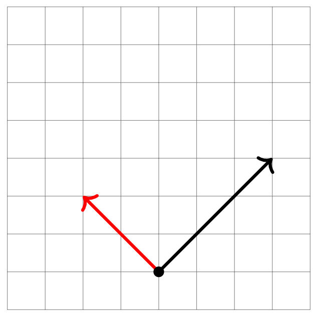
The following plots give examples of aligned vectors, anti-aligned vectors, vectors that make an acute angle, vectors that make an obtuse angle, and orthogonal vectors.











Properties of the Dot Product

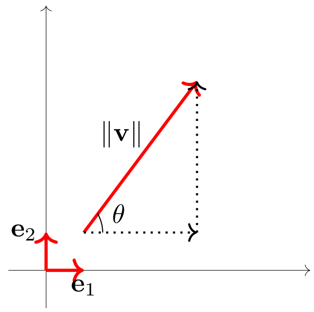
Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors in $\mathbb{R}^n$, c be a scalar, and θ is the angle betwen $\mathbf{u}$ and $\mathbf{v}$.

1. $\mathbf{u}^T\mathbf{u} = \|\mathbf{u}\|^2$
2. $\mathbf{u}^T\mathbf{v} = \mathbf{v}^T\mathbf{u}$
3. $\mathbf{u}^T(\mathbf{v}+\mathbf{w}) = \mathbf{u}^T\mathbf{w} + \mathbf{v}^T\mathbf{w}$
4. $c(\mathbf{u}^T\mathbf{v}) = c(\mathbf{u}^T\mathbf{v})$
5. $\mathbf{0}^T\mathbf{u} = 0$
6. $\mathbf{u}^T\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$
7. $\mathbf{u}^T\mathbf{v} = 0 \iff \mathbf{u} \perp \mathbf{v}$

Example 1.28 Vector reeprepresentations

We can use 3 equivalent ways to denote vectors in $\mathbb{R}^2$:

1. $\mathbf{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$: component notation. The vector is written as a column surrounded by (or between) square brackets. The numbers are called the components or coordinates of the vector, for example, $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$.
2. $\mathbf{v} = v_x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + v_y \begin{bmatrix} 0 \\ 1 \end{bmatrix} = v_x\mathbf{e}_1 + v_y\mathbf{e}_2$: unit vector notation. The vector is expressed as a combination of the unit vectors, for example, $\begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, or $3\mathbf{e}_1 + 4\mathbf{e}_2$.
3. $\mathbf{v} = \|\mathbf{v}\|\angle\theta$: length-and-direction notation, also known as polar coordinates. The vector is expressed in terms of its length $\|\mathbf{v}\|$ and the angle θ that the vector makes with the x-axis, for example, $5\angle\theta$. $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3\mathbf{e}_1 + 4\mathbf{e}_2 = 5\angle 30°$ is as in



$\|\mathbf{v}\|$
θ
$\mathbf{e}_2$
$\mathbf{e}_1$

2

Matrices and Linear Transformations

2.1 Matrices and Matrix operations

A Matrix as a Collection of Vectors

The 3×4 matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 0 & 7 & 8 \\ 0 & 10 & 11 & 12 \end{bmatrix}$$

may be represented as

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 \end{bmatrix}$$

where

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix}, \ \mathbf{a}_2 = \begin{bmatrix} 2 \\ 0 \\ 10 \end{bmatrix}, \ \mathbf{a}_3 = \begin{bmatrix} 3 \\ 7 \\ 11 \end{bmatrix}, \ \mathbf{a}_4 = \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix}$$

We say this 3×4 matrix above is a collection of 4 vectors in $\mathbb{R}^3$.

Example 2.1 Represent a linear system in a matrix form and an augmented matrix form

If we have a linear system of 3 equations in 3 unknowns:

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 2 \\ -x_1 + 4x_2 - 5x_3 &= 7 \\ x_2 - 9x_3 &= 11 \end{aligned}$$

we can write the coefficients on the left-hand side into a matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & -5 \\ 0 & 1 & -9 \end{bmatrix}$$

We can also add the right-hand side constants into the coefficient matrix as the last column, and we call it the augmented matrix of a linear system:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ -1 & 4 & -5 & 7 \\ 0 & 1 & -9 & 11 \end{array}\right]$$

We can also write it in a matrix form ($\mathbf{Ax} = \mathbf{b}$):

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & -5 \\ 0 & 1 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 11 \end{bmatrix}$$

Matrix Operations

Matrix Addition

Matrix addition is element wise. You can add (or subtract) matrices by adding (or subtracting) corresponding entries if they are of the same size.

Example 2.2 An example of matrix addition.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} + \begin{bmatrix} -1 & 0 & -2 \\ 0 & -3 & -4 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 1 \\ 4 & 2 & 2 \end{bmatrix}$$

Scalar Multiplication

You can multiply a matrix by a number by multiplying each entry by the number.

Example 2.3 An example of scalar multiplication

$$7 \cdot \begin{bmatrix} 1 & 1 \\ 3 & 0 \\ -2 & 13 \end{bmatrix} = \begin{bmatrix} 7 & 7 \\ 21 & 0 \\ -14 & 91 \end{bmatrix}$$

Example 2.4 Find the values a and b such that $\begin{bmatrix} a & -2 \\ -3 & -b \end{bmatrix} + \begin{bmatrix} -2a & 2 \\ 3 & 3b \end{bmatrix} = 2\mathbf{I}$.

Two matrices are equal if they have the same number of rows and the same number of columns, and precisely the same entries in the same locations.

The $n \times n$ matrix $\mathbf{I}_n$ is called the identity matrix whose entries are zero except for the diagonal entries, which are 1. For instance,

$$\mathbf{I}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

We compute

$$\begin{bmatrix} a & -2 \\ -3 & -b \end{bmatrix} + \begin{bmatrix} -2a & 2 \\ 3 & 3b \end{bmatrix} = \begin{bmatrix} -a & 0 \\ 0 & 2b \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Thus, $a = -2$, $b = 1$

Matrix Multiplication

"Row times column" is an easy way to compute the product of two matrices. If $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is an $n \times p$ matrix, then the product $\mathbf{AB} = \mathbf{C}$ is an $m \times p$ matrix. We use c_{ij} to denote the entry in row i and column j of matrix $\mathbf{C}$. c_{ij} equals the dot product of row i of $\mathbf{A}$ and column j of $\mathbf{B}$:

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{in}b_{nj} \sum_{k=1}^{n} a_{ik}b_{kj}$$

for $i = 1$, ..., m and $j = 1, ..., p$

Example 2.5 Compute the product of a matrix $\mathbf{A}$ and a 2-dimensional vector $\mathbf{x}$

$$\mathbf{AB} = \begin{bmatrix} -1 & 2 \\ 0 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} -1 \times 3 + 2 \times 5 \\ 2 \times 5 \\ 3 \times 3 + 1 \times 5 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \\ 14 \end{bmatrix}$$

Example 2.6 Compute the product of $\mathbf{AB}$.

$$\mathbf{AB} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1 \times 5 + 2 \times 7 & 1 \times 6 + 2 \times 8 \\ 3 \times 5 + 4 \times 7 & 3 \times 6 + 4 \times 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 44 \end{bmatrix}$$

Transpose of A Matrix

If $\mathbf{A}$ is an $m \times n$ matrix, the transpose $\mathbf{A}^T$ of $\mathbf{A}$ is obtained by swapping the rows and columns of $\mathbf{A}$. For example,

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

The transpose of an $m \times n$ matrix is an $n \times m$ matrix.

Example 2.7 Symmetric matrix

A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is said to be symmetric if $\mathbf{A} = \mathbf{A}^T$. For instance,

$$\mathbf{A} = \begin{bmatrix} 1 & 7 & 3 \\ 7 & 4 & 5 \\ 3 & 5 & 2 \end{bmatrix}$$

is symmetric. The rank of a symmetric matrix $\mathbf{A}$ is equal to the number of non-zero eigenvalues of $\mathbf{A}$.

Inverse of A Matrix

The inverse of an $m \times n$ matrix ia a matrix $\mathbf{A}^{-1}$ which satisfies

$$\mathbf{AA}^{-1} = \mathbf{I} \text{ and } \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

An $n \times n$ matrix which does not have an inverse is called **singular**.

If $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\mathbf{A}$ is invertible, then the inverse of $\mathbf{A}$ is

$$\mathbf{A}^{-1} = (ad - bc)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The number $ad - bc$ is called the **determinant** of the matrix.

2.2 Matrices and Linear Transformations

We use the "row-times-column" to compute a matrix product $\mathbf{AB}$:

$$\mathbf{AB} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 2 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & 4 \end{bmatrix}$$

If $\mathbf{B}$ has only one column, then it becomes a matrix-vector multiplication. For instance,

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

If $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, we have

$$\mathbf{Ax} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

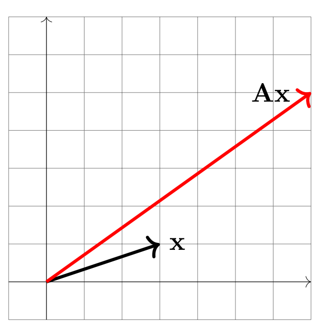
We can think of the product $\mathbf{Ax}$ as a linear combination of the columns of $\mathbf{A}$ with weights given by the entries of $\mathbf{x}$.

We can think of each column of the product $\mathbf{AB}$ as a linear combination of the columns of $\mathbf{A}$ with weights given by the corresponding column of $\mathbf{B}$.

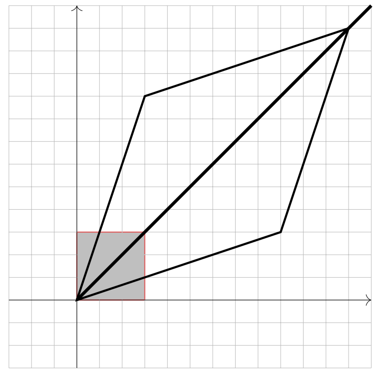
And the fundamental way of viewing a matrix is as a function that transforms one vector to another.

Example 2.8 Matrix $\mathbf{A}$ maps one vector $\mathbf{x}$ to $\mathbf{Ax}$

$$\mathbf{Ax} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$$



Example 2.9 The transformation $\mathbf{A}$



We write the linear transformation for shearing:

$$T(\begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3x + y \\ x + 3y \end{bmatrix}$$

The matrix of T is

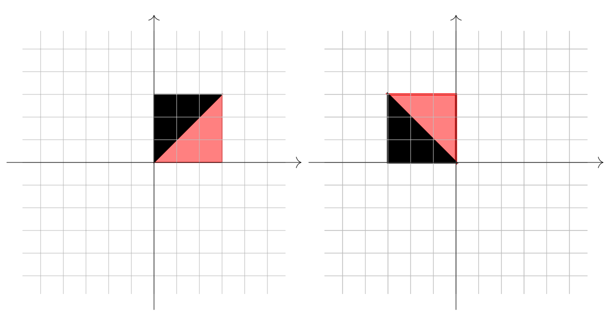
$$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

The black lines are eigenvectors.

Example 2.10 Matrix of $\mathbf{T} : \mathbb{R}^2 \to \mathbb{R}^2$: counter clockwise $90°$ rotation

Interpreting a matrix $\mathbf{A}$ in terms of a function, or an operator, effecting the transformation $x \longmapsto \mathbf{A}\mathbf{x}$, is the fundamental idea in linear algebra.

Here's the before-and-after plot:



We write the linear transformation for counter clockwise $90°$:

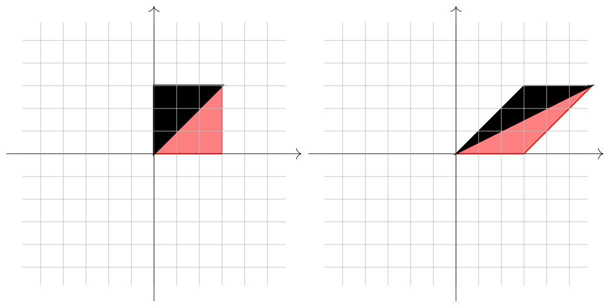
$$T(\begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

The matrix of T is

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Example 2.11 Matrix of $T : \mathbb{R}^2 \to \mathbb{R}^2$: Shearing

It keeps one line (the x-axis) fixed while shifting all other points by varing distances along lines that are parallel to the x-axis. Here's the before-and-after plot:



We write the linear transformation for shearing:

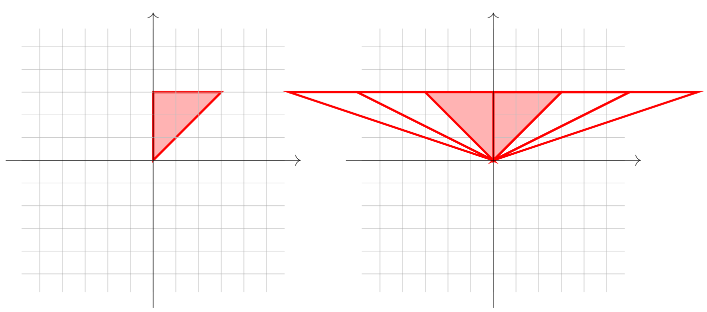
$$T(\begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+y \\ y \end{bmatrix}$$

The matrix of T is

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

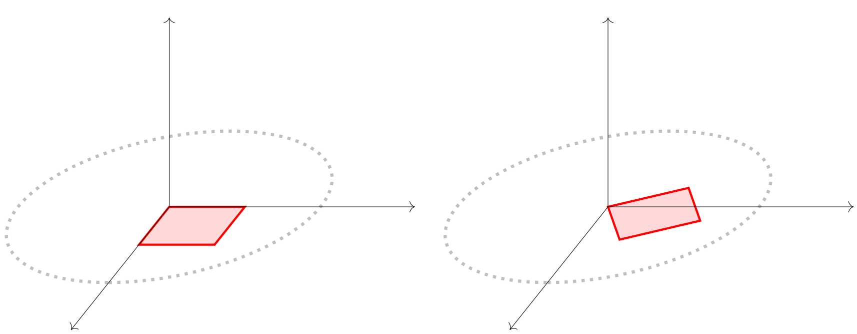
Example 2.12 Shear inverse transformation

Here's the before-and-after plot:



Matrix transformation: $T(\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+ay \\ y \end{bmatrix})$, induced by matrix $\mathbf{A} = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$

Example 2.13 A rotation in 3-dimensional space: the rotation is about the z-axis, so all the action is taking place in the xy-plane.



Properties of Matrix Operations

The operations are as follows:

- Addition: If $\mathbf{A}$ and $\mathbf{B}$ are matrices of the same size $m \times n$, then $\mathbf{A} + \mathbf{B}$, their sum, is a matrix of size $m \times n$.
- Multiplication by scalars: If $\mathbf{A}$ is a matrix of size $m \times n$, and c is a scalar, then $c\mathbf{A}$ is a matrix of size $m \times n$.
- Matrix multiplication: If $\mathbf{A}$ is a matrix of size $m \times n$ and $\mathbf{B}$ is a matrix of size $n \times p$, then the product $\mathbf{AB}$ is a matrix of size $m \times p$.
- Vectors: A vector of length n can be treated as a matrix of size $n \times 1$, and the operation of vector addition, multiplication by scalars, and mutliplying a matrix by a vector agree with the corresponding matrix operations.
- Transpose: If $\mathbf{A}$ is a matrix of size $m \times n$, then its transpose $\mathbf{A}^T$ is a matrix of size $m \times n$.
- Identity matrix: $\mathbf{I}_n$ is the $n \times n$ identity matrix; its diagonal elements are equal to 1 and its offdiagonal elements are equal to 0.
- Zero matrix: We denote by $\mathbf{0}$ the matrix of all zeros of relevant size.
- Inverse: If $\mathbf{A}$ is a square matrix, then its inverse $\mathbf{A}^{-1}$ is a matrix of the same size. Not every square matrix has an inverse. The matrices that have inverses are called invertible.

Assuming that r, s are scalars and the sizes of the matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ are chosen so that each operation is well defined, the properties of these operations are:

1. $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
2. $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$
3. $\mathbf{A} + \mathbf{0} = \mathbf{A}$
4. $r(\mathbf{A} + \mathbf{B}) = r\mathbf{A} + r\mathbf{B}$
5. $(r + s)\mathbf{A} = r\mathbf{A} + s\mathbf{A}$
6. $r(s\mathbf{A}) = (rs)\mathbf{A}$
7. $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$
8. $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$
9. $(\mathbf{B} + \mathbf{C})\mathbf{A} = \mathbf{BA} + \mathbf{CA}$
10. $r(\mathbf{AB}) = (r\mathbf{A})\mathbf{B} = \mathbf{A}(r\mathbf{B})$
11. $\mathbf{I}_m\mathbf{A} = \mathbf{A} = \mathbf{A}\mathbf{I}_n$
12. $(\mathbf{A}^T)^T = \mathbf{A}$
13. $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$
14. $(r\mathbf{A})^T = r\mathbf{A}^T$
15. $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$
16. $(\mathbf{I}_n)^T = \mathbf{I}_n$
17. $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$
18. $(r\mathbf{A})^{-1} = r^{-1}\mathbf{A}^{-1}, r \neq 0$
19. $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
20. $(\mathbf{I}_n)^{-1} = \mathbf{I}_n$
21. $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$
22. $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$

2.3 Gaussian Elimination

Gaussian Elimination

Three row operations you can do with the augmented matrix of a linear system are:

1. swapping two rows
2. multiplying a row by a nonzero number
3. adding a multiple of one row to another

Example 2.14 Solve a linear system of 2 equations in 2 unknowns.

Consider the system of equations:

$$\begin{aligned} 2x_1 - x_2 &= 0 \\ 3x_1 - x_2 &= 3 \end{aligned}$$

We can write in augmented matrix form:

$$\left[\begin{array}{cc|c} 2 & -1 & 0 \\ 3 & -1 & 3 \end{array}\right]$$

We can multiply the first row (R1) by $-\frac{3}{2}$ and add it to the 2nd row (R2), etc...

$$\left[\begin{array}{cc|c} 2 & -1 & 0 \\ 3 & -1 & 3 \end{array}\right] \to \left[\begin{array}{cc|c} 2 & -1 & 0 \\ 0 & \frac{1}{2} & 3 \end{array}\right]$$

Back substitution yields: $\frac{1}{2}x_2 = 3$, and $2x_1 - x_2 = 0$.

Thus, $x_1 = 3$, $x_2 = 6$

This procedure is called Gaussian elimination.

Example 2.15 Solve a linear system of 3 equations in 3 unknowns.

Consider the system of equations:

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 3 \\ 4x_1 + 3x_2 + 3x_3 &= 9 \\ 8x_1 + 7x_2 + 9x_3 &= 25 \end{aligned}$$

The linear syste can be written in a matrix notation:

$$\mathbf{Ax} = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{bmatrix} 3 \\ 9 \\ 25 \end{bmatrix}$$

or in an augmented matrix notation:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 3 \\ 4 & 3 & 3 & 9 \\ 8 & 7 & 9 & 25 \end{array}\right]$$

We use Gaussian elimination:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 3 \\ 4 & 3 & 3 & 9 \\ 8 & 7 & 9 & 25 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 1 & 3 \\ 0 & 1 & 1 & 3 \\ 0 & 3 & 5 & 13 \end{array}\right]$$
$$\rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 1 & 3 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 2 & 4 \end{array}\right]$$

Back substitution yields:

$$2x_3 = 4,\ x_2 + x_3 = 3,\ \text{and}\ 2x_1 + x_2 + x_3 = 3$$

Thus, $x_1 = 0$, $x_2 = 1$, $x_3 = 2$

Quick Summary: Gaussian elimination proceeds in steps until an upper triangular matrix is obtained for back-substitution.

Example 2.16 Solutions of linear systems represented in Augmented Matrix

(i) **No solutions**:

$$\begin{aligned} x_1 + 2x_2 &= 1 \\ 2x_1 + 4x_2 &= 3 \end{aligned}$$

We represent the linear system into the augmented matrix form and then perform Gaussian elimination on it:

$$\left[\begin{array}{cc|c} 1 & 2 & 1 \\ 2 & 4 & 3 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 0 & 1 \end{array}\right]$$

The last row has the form $\left[\begin{array}{cc|c} 0 & 0 & 1 \end{array}\right]$, which is a contradiction; therefore, the linear system has no solution. This is a 2×2 linear system case but this applies to any $m \times n$ linear systems.

(ii) **Infinitely many solutions**:

If we change the output vector to $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, the linear system will have infinitely many solutions.

$$\begin{aligned} x_1 + 2x_2 &= 1 \\ 2x_1 + 4x_2 &= 2 \end{aligned}$$

$$\left[\begin{array}{cc|c} 1 & 2 & 1 \\ 2 & 4 & 2 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 0 & 0 \end{array}\right]$$

Thus, this linear system above has a free variable, and therefore, infinitely many solutions.

(iii) **Unique solution**:

$$\begin{aligned} x_1 + 2x_2 &= 6 \\ 2x_1 + x_2 &= 6 \end{aligned}$$

$$\left[\begin{array}{cc|c} 1 & 2 & 6 \\ 2 & 1 & 6 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 6 \\ 0 & -3 & -6 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 2 \end{array}\right]$$

Thus, x_1=2, x_2=2.

LU Factorization

LU factorization is essentially the matrix representation of Gaussian elimination.

Elimination Matrices

An elimination matrix is the one obtained by performing a single elementary row operation on an identity matrix. If the elimination matrix used to eliminate the entry in row m and column n is denoted $\mathbf{E}_{ij}$. Elimination matrices are considered linear combinations because they can be expressed as a sum of scalar multiples of the identity matrix, where the scalars represent the coefficients used to perform row operations during the Gaussian elimination.

Example 2.17 Elimination matrices and inverses of elimination matrices

1. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

2. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 7 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 7a+g & 7b+h & 7c+i \end{bmatrix}$

3. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$

Example 2.18 An example of **LU** factorization

Consider the system of equations

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 2 \\ 2x_1 + 6x_2 + x_3 &= 7 \\ x_1 + x_2 + 4x_3 &= 3 \end{aligned}$$

We write the linear system in a matrix-vector form

$$\mathbf{Ax} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 3 \end{bmatrix} = \mathbf{b}$$

We eliminate the subdiagonal entries of the first column of $\mathbf{A}$:

$$\mathbf{E}_{21}\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{bmatrix}$$

$$\mathbf{E}_{31}\mathbf{E}_{21}\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

Next step would be to eliminate teh third entries from the second column which will result an upper triangular matrix $\mathbf{U}$:

$$\mathbf{E}_{32}\mathbf{E}_{31}\mathbf{E}_{21}\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ - & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & \frac{5}{2} \end{bmatrix} = \mathbf{U}$$

We have

$$\begin{aligned} \mathbf{L} = (\mathbf{E}_{32}\mathbf{E}_{31}\mathbf{E}_{21})^{-1} &= \mathbf{E}_{21}^{-1}\mathbf{E}_{31}^{-1}\mathbf{E}_{32}^{-1} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -\frac{1}{2} & 1 \end{bmatrix} \end{aligned}$$

Now we have

$$\mathbf{A} = \mathbf{L}\mathbf{U} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & \frac{5}{2} \end{bmatrix}$$

Permutation

A permutation matrix $\mathbf{P}$ is constructed by swapping rows of the identity matrix. Multiplication by a permutation matrix $\mathbf{P}$ swaps the rows of a matrix. When performing Gaussian elimination, we use permutation matrices to move zeros out of pivot positions. Then we can factorize $\mathbf{PA} = \mathbf{LU}$. $\mathbf{P}^{-1} = \mathbf{P}^T$, i.e., $\mathbf{P}^T\mathbf{P} = \mathbf{I}$. For example,

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

permutes $\mathbf{v}$ as

$$\mathbf{P} \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_3 \\ \mathbf{v}_2 \end{bmatrix}$$

Example 2.19 An example of $\mathbf{PA} = \mathbf{LU}$

Consider

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 2 & -1 & -1 \end{bmatrix}$$

Applying Gaussian elimination, we have

$$\mathbf{E}_{31}\mathbf{E}_{21}\mathbf{A} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 2 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -2 \\ 0 & -3 & -3 \end{bmatrix}$$

does not have an $\mathbf{LU}$ factorization. However,

$$\mathbf{PA} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 2 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & -1 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -3 \\ 0 & 0 & -2 \end{bmatrix} = \mathbf{LU}$$

Gauss-Jordan Elimination

The algorithm to compute the inverse of a nonsingular matrix is Gauss-Jordan Elimination.

Example 2.20 Find the inverse of

$$\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$$

We form the large augmented Matrix:

$$\begin{bmatrix} 1 & 3 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{bmatrix}$$

Applying row operations to get the inverse:

$$\begin{bmatrix} 1 & 3 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & -5 & -3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & \frac{3}{5} & \frac{-1}{5} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & \frac{-4}{5} & \frac{3}{5} \\ 0 & 1 & \frac{3}{5} & \frac{-1}{5} \end{bmatrix}$$

Thus, $\mathbf{A}^{-1} = \begin{bmatrix} -\frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{-1}{5} \end{bmatrix}$

Real-world problems deal with large linear systems, and numerical solution schemes for weater prediction, image and video processing, often require dealing with matrices with more than a million entries. Such systems can only be worked out with the sophisticated supercomputer. It is essential that we understand the computational details of computing methods.

2.4 Subspaces

Vector Space

A vector space is defined by a set $\mathbf{V}$ which is closed under two basic operations on $\mathbf{V}$, vector addition and scalar mutliplication,

1. $\mathbf{x}, \mathbf{y} \in \mathbf{V} \Rightarrow \mathbf{x} + \mathbf{y} \in \mathbf{V}$
2. $\mathbf{x} \in \mathbf{V}, a \in \mathbb{R} \Rightarrow a\mathbf{x} \in \mathbf{V}$

We can add vectors and multiply vectors by numbers, which means that we can work with linear combinations of vectors, which follow the rules of a vector space.

$\mathbb{R}^n$: n-dimensional Vector Space

$\mathbb{R}^n$ is a vector space consisting of the set of vectors

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

where $x_i \in \mathbb{R}$. In $\mathbb{R}^n$, the basic operations are definded by component-wise addition and multiplication, such that

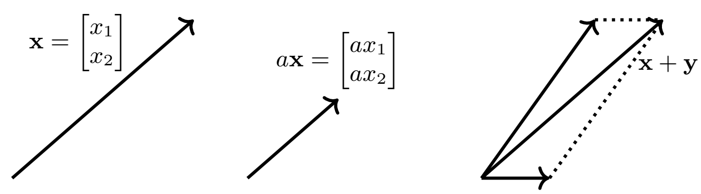
(i)

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

(ii)

$$a\mathbf{x} = \begin{bmatrix} ax_1 \\ ax_2 \\ \vdots \\ ax_n \end{bmatrix}$$

Geometrically, $\mathbb{R}^n$ can be interpreted as the directed line segments, defined by both (i) direction with respect to the origin, and (ii) magnitude (or length).



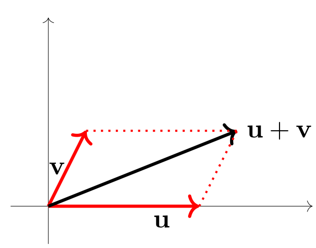
Subspaces

A vector space that is contained inside of another vector space is called a subspace. A subspace of $\mathbb{R}^n$ is a subset W of $\mathbb{R}^n$ satisfying:

1. Non-emptiness: The zero vector is in W.
2. Closure under addition: If $\mathbf{u}$ and $\mathbf{v}$ are in W, then $\mathbf{u} + \mathbf{v}$ is also in W.
3. Closure under scalar multiplication: If $\mathbf{v}$ is in W and λ is in $\mathbb{R}$, then $\lambda\mathbf{v}$ is also in W.

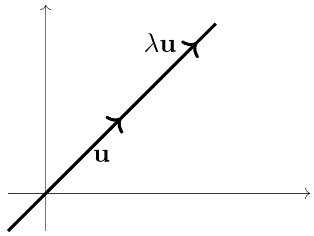
Visualizing Closure Under Addition

If $\mathbf{u}$ and $\mathbf{v}$ are in W, then so is $\mathbf{u} + \mathbf{v}$:



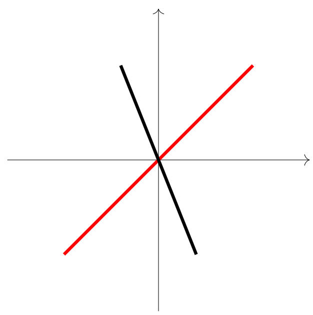
Visualizing Closure Under Scalar Multiplication

If $\mathbf{v}$ is in W and λ is in R, then $\lambda\mathbf{v}$ is also in W:



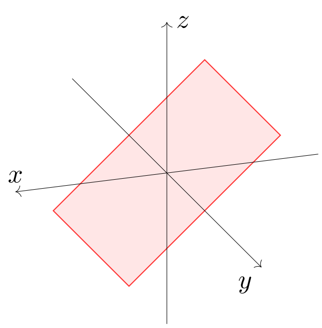
Subspaces of $\mathbb{R}^2$

$\mathbb{R}^2$ is the entire xy plane. The subspaces of $\mathbb{R}^2$ are $\{0\}$, $\mathbb{R}^2$, and lines passing through the origin. For example, the set of all vectors $c\mathbf{v}$, where c is a real number, forms a subspace of $\mathbb{R}^2$.



Subspaces of $\mathbb{R}^3$

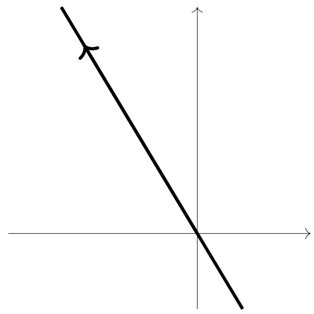
In $\mathbb{R}^3$, the subspaces are $\{0\}$, $\mathbb{R}^3$, and lines or planes passing through the origin.



Example 2.21 A line in $\mathbb{R}^2$ through the origin is a subspace.

$$5x + 3y = 0$$

This line is closed under addition and scalar multiplication. Additionally, this line contains the origin. Thus, $\left\{ \begin{bmatrix} x \\ y \end{bmatrix} : 5x + 3y = 0 \right\}$ is a subspace spanned by the single vector $\begin{bmatrix} -3 \\ 5 \end{bmatrix}$.



Example 2.22 A plane $\mathbf{P}$ in $\mathbb{R}^3$ thorugh the origin is a subspace.

$$ax + by + cz = 0$$

This plane can be expressed as

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$\mathbf{P}$ is closed under addition and scalar multiplication. Additionally, $\mathbf{P}$ contains the origin. Thus, $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : \; ax + by + zx = 0 \right\}$ is a subspace.

Span

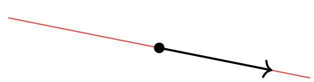
The span of the set of m-dimensional vectors $\{\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_n\}$ is the set in $\mathbb{R}^m$ of all possible linear combinations of the vectors $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_n$. We use the notation

$$\text{span}\{\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_n\} \in \mathbb{R}^m$$

The span of any set of nonzero vectors, span$\{\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_n\}$ is always an infinite collection of vectors.

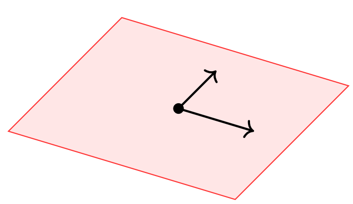
Span of a Single Vector

The span of a single nonzero vector in $\mathbb{R}^2$ or $\mathbb{R}^3$ is a line. All possible linear combinations of a single vector $\mathbf{v}$ are vectors of the form $\lambda\mathbf{v}$, as in



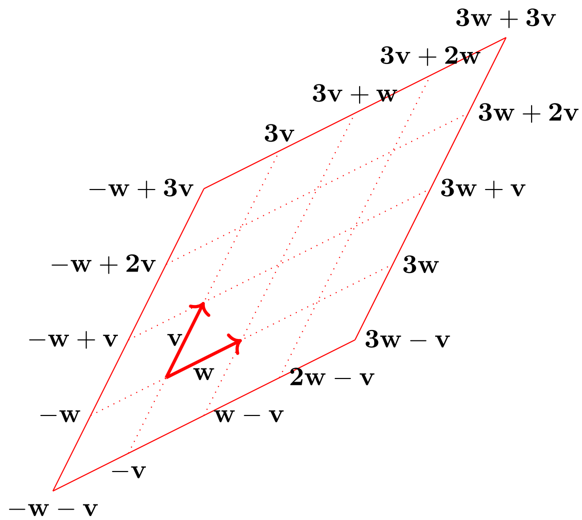
Span of A Set of Two Noncollinear Vectors

The span of a set of two noncollinear vectors belong to a plane. The span of two vectors that point in different directions is the plane containing both vectors, as in



Example 2.23 Span of two noncollinear vectors

For two vectors, the linear combinations are $c\mathbf{v} + d\mathbf{w}$. For two noncollinear vectors, all linear combinations of $\mathbf{v}$ and $\mathbf{w}$ fill a plane containing the origin and both vectors.



2.5 Solving $\mathbf{Ax} = \mathbf{b}$

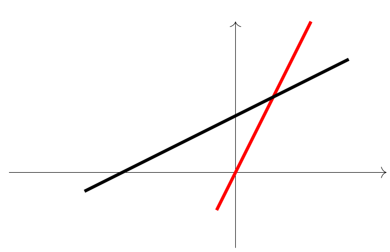
Geometry of Linear Systems

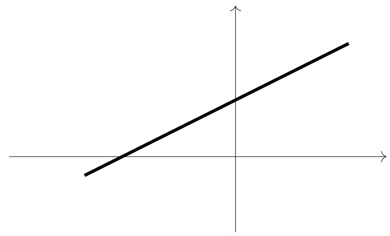
Solutions to 2×2 Linear Systems

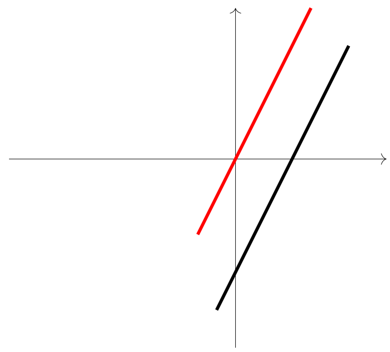
A $m \times n$ linear system is inconsistent if it has no solutions; it's consistent otherwise. Solutions of 2×2 linear systems fall into one of the following three cases:

1. There exists a unique solution.
2. There exists infinitely many solutions.
3. There exists no solution.

The solutions of each equation in a 2×2 linear system represents a line in $\mathbb{R}^2$. Two lines in $\mathbf{R}^2$ can intersect at a point, or can be coincident, or can be parallel but no coincident respectively:

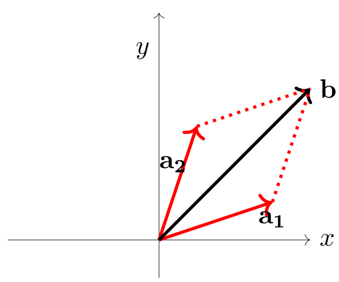


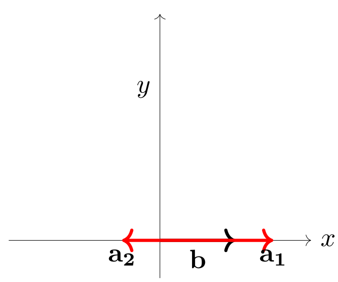


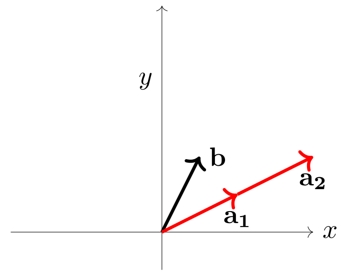


Solutions of 2×2 linear systems as vector equations

The 2×2 linear system $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = \mathbf{b}$ has a solution if and only if the output vector $\mathbf{b}$ belongs to the span$\{\mathbf{a}_1, \mathbf{a}_2\}$, i.e., $\mathbf{b}$ $\in$span$\{\mathbf{a}_1, \mathbf{a}_2\}$. The 2×2 linear system $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = \mathbf{b}$ is consistent (having unique solution or infinitely many solutions) if $\mathbf{b}$ $\in$span$\{\mathbf{a}_1, \mathbf{a}_2\}$, while it has no solution if $\mathbf{b}$ $\notin$span$\{\mathbf{a}_1, \mathbf{a}_2\}$. Examples of solutions to 2×2 linear systems having a unique, infinitely many, and no solution, represented in the following plots respectively:







Null Space

The null space of an $m \times n$ matrix $\mathbf{A}$ is the set of all solutions of the homogeneous linear system $\mathbf{Ax} = \mathbf{0}$. The null space of an $m \times n$ matrix $\mathbf{A}$ is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions to a system $\mathbf{Ax} = \mathbf{0}$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

Example 2.24 Find the solution to an homogeneous linear system

Consider the linear system

$$\begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

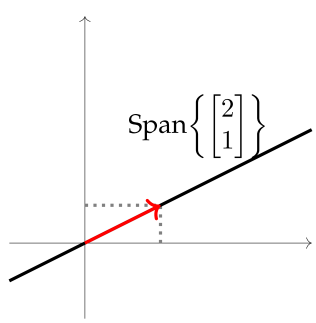
The solution can be found by using row reduction and then back substitution, as in

$$\begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} \to \begin{bmatrix} 2 & -4 \\ 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 & = 2x_2 \\ x_2 & \text{free variable} \end{cases}$$

Thus, the set of all solutions to the linear system above is given by

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \; \forall x_2 \in \mathbb{R} \; \Rightarrow \mathbf{x}_h \in \text{span}\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$$

The null space is a line in $\mathbb{R}^2$.



Column Space

The column space of an $m \times n$ matrix $\mathbf{A}$ is the subspace of $\mathbb{R}^m$ spanned by the columns of $\mathbf{A}$. It is written as Col($\mathbf{A}$). The column space of $\mathbf{A}$ consists of all linear combinations of the columns of $\mathbf{A}$, i.e., the span of the columns of $\mathbf{A}$. The columns space is defined to be a span, so it is a subspace. Since the column space of $\mathbf{A}$ is described as the span of the columns of $\mathbf{A}$, columns of $\mathbf{A}$ are vectors in $\mathbb{R}^m$, which means all linear combinations of the columns are also in $\mathbb{R}^m$. We see that Col($\mathbf{A}$) is a subspace of $\mathbb{R}^m$.

Example 2.25 Col($\mathbf{A}$)

$$\mathbf{A} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 9 \\ 1 & 7 & 8 \end{bmatrix}$$

$$\text{Col}(\mathbf{A}) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$$

$\mathbf{v}_3$ can be written as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. Thus, Col($\mathbf{A}$) is 2-dimensional subspace of $\mathbb{R}^3$ and therefore a plane.

Complete Solutions to Linear Systems

A complete solution to a linear system refers to a representation of all possible solutions ot the system, including any parameters or free variables that might exist. We get the complete solution by adding the particular solution to all the vectors in the nullspace.

Example 2.26 Solve a linear system of 1 equation in 2 unknowns.

Consider the linear system:

$$x_1 + 2x_2 = 3$$

The solution set to this equation can be parameterized by setting $x_2 = t$ and $x_1 = 3 - 2t$. In other words, the solution set is $\{(3-2t, t) \mid t \in \mathbb{R}\}$, as in

Thus, $\left\{ \begin{array}{ll} x_1 & = 3 - 2x_2 \\ x_2 & \text{free variable} \end{array} \right.$ is the solution.

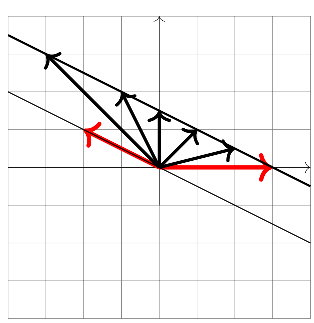
It can also be expressed as

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 - 2x_2 \\ x_2 \end{bmatrix} \Rightarrow \mathbf{x} = x_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

The solution above can also be expressed as

$$\mathbf{x} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} + x_h,\ x_h \in \text{span}\left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$$

We represent the solution on the plane, as in



Example 2.27 Solve a linear system of 3 equations in 4 uknowns.

Consider the system

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 1 \\ 2x_1 + 2x_2 - x_3 + x_4 &= 2 \\ 3x_1 + 3x_2 + x_3 + x_4 &= 4 \end{aligned}$$

We begin to write down the augmented matrix and then use row reduction.

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & -1 & 1 & 2 \\ 3 & 3 & 1 & 1 & 4 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & -3 & -1 & 0 \\ 0 & 0 & -2 & -2 & 1 \end{array}\right]$$
$$\to \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 & \frac{-3}{4} \end{array}\right]$$
$$\Rightarrow \begin{cases} x_1 + x_2 + x_3 + \phantom{\frac{1}{3}}x_4 = 1 \\ \qquad\qquad x_3 + \frac{1}{3}x_4 = 0 \\ \qquad\qquad\qquad\quad x_4 = -\frac{3}{4} \end{cases}$$

By using t as a parameter for the variable x_2, our solution set is $\{-t+\frac{3}{2},\ t,\ \frac{1}{4,},\ \frac{-3}{4} \mid t \in \mathbb{R}\}$. x_2 is called a free variable and t is called its parameter, while x_1, x_3 and x_4 are called determined variables.

Columns 1, 3 and 4 are pivot columns, and x_2 is a free variable.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -x_2 + \frac{3}{2} \\ x_2 \\ \frac{1}{4} \\ \frac{-3}{4} \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{3}{2} \\ 0 \\ \frac{1}{4} \\ \frac{-3}{4} \end{bmatrix}, \forall x_2 \in \mathbb{R} \Rightarrow \mathbf{x}_h \in \text{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

The null space is a line in $\mathbb{R}^4$.

2.6 Change of Basis

Linear Independence

A non-empty set of m-dimensional vectors $\{\mathbf{v}_1, ..., \mathbf{v}_n\}$ is linearly independent if the $m \times n$ homogeneous linear system $\mathbf{V}\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$, where the matrix $\mathbf{V}$ is given by $\mathbf{V} = [\mathbf{v}_1\mathbf{v}_2...\mathbf{v}_n]$ and the vector $\mathbf{x} \in \mathbb{R}^n$. The set of vectors above is called linearly depedent if there exists a non-trivial solution, $\mathbf{x} \neq \mathbf{0}$ of the linear system $\mathbf{V}\mathbf{x} = \mathbf{0}$.

Example 2.28 Is the set of vectors linearly independent?

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \ \mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \ \mathbf{v}_3 = \begin{bmatrix} 8 \\ 10 \\ 12 \end{bmatrix}$$

This is equivalently asking if the homogeneous linear system has a nontrivial solution. we form a coefficient matrix and then use Gaussian elimination:

$$\begin{bmatrix} 1 & 4 & 8 \\ 2 & 5 & 10 \\ 3 & 6 & 12 \end{bmatrix} \to \begin{bmatrix} 1 & 4 & 8 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \to \begin{bmatrix} 1 & 4 & 8 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

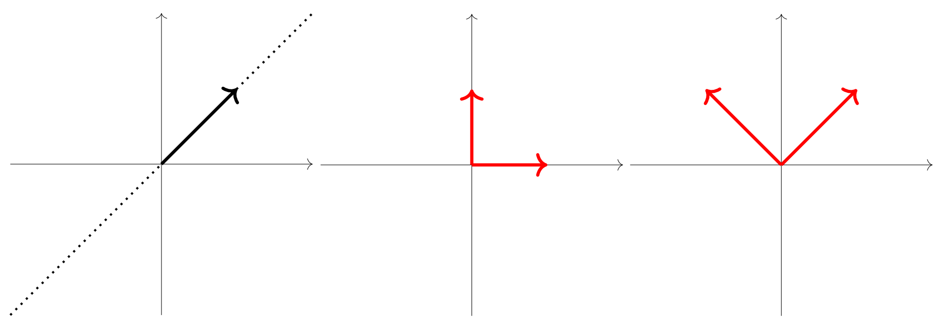
There exists nontrivial solutions; thus, they are linearly dependent.

Bases

A set of vectors $\{\mathbf{v}_1, ..., \mathbf{v}_n\}$ in $\mathbb{R}^m$ is called a basis for $\mathbb{R}^m$ if the set of vectors spans $\mathbb{R}^m$ and is linearly indepedent. The basis of a vector space tells us everything we need to know about that vector space.

Example 2.29 Set of vectors that span (via linear combination) all points in a vector space

The 1-dimensional vector space (subspace of $\mathbb{R}^2$), two different orthonormal bases for the same 2-dimensional vector space can be visualized respectively:



Example 2.30 One basis for $\mathbb{R}^2$ is $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$. These are independent because

$$c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

is only possible when $c_1 = c_2 = 0$. These vectors span $\mathbb{R}^2$.

Example 2.31 The vectors $\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 6 \\ 6 \\ 14 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 7 \end{bmatrix} \right\}$ cannot form a basis for $\mathbb{R}^3$.

These vectors cannot form a basis because the they are not linearly independent.

Example 2.32 Standard basis for $\mathbb{R}^3$

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{e}_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Every vector in $\mathbb{R}^3$ can be written as a linear combination of $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$, with unique coefficients. For instance,

$$\begin{bmatrix} 6 \\ 7 \\ 9 \end{bmatrix} = 6\mathbf{e}_1 + 7\mathbf{e}_2 + 9\mathbf{e}_3$$

Change of Basis

Any vector can be decomposed in an unique way in terms of a basis. Let $\mathbf{V}$ be an n-dimensional vector space, and $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a basis of $\mathbf{V}$. Then each vector $\mathbf{x} \in \mathbf{V}$ has a unique decomposition:

$$\mathbf{x} = c_1\mathbf{v}_1 + \ldots + c_n\mathbf{v}_n$$

where the n scalars c_i are called components or coordinates of $\mathbf{x}$ with respect to this basis, written as

$$[\mathbf{x}]_v = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$$

.

Example 2.33 Find the components of $\mathbf{x} = 2\mathbf{e}_+ 4\mathbf{e}_2$ in the basis $\{\mathbf{u}_1, \mathbf{u}_2\}$.

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \ \mathbf{u}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{x} = 2\mathbf{e}_1 + 4\mathbf{e}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}_e \Rightarrow [\mathbf{x}]_e = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

The vectors $\mathbf{u}_1$, $\mathbf{u}_2$ from a basis so there exists constants c_1, c_2 such that

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

To solve the augmented matrix

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & 2 \end{bmatrix}$$

Thus, $c_1 = 3, c_2 = -1$, then $[\mathbf{x}]_u = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

2.7 Eigenvalues and Eigenvectors

Definition

A matrix $\mathbf{A}$ acts on a vector $\mathbf{x}$ like a function does, with input $\mathbf{x}$ and output $\mathbf{Ax}$. In general, the square matrix $\mathbf{A}$ differs from $\mathbf{x}$ in both magnitude and direction. However, in some special cases where $\mathbf{x}$ is an eigenvector of $\mathbf{A}$, multiplied by $\mathbf{A}$ leaves its direction unchanged.

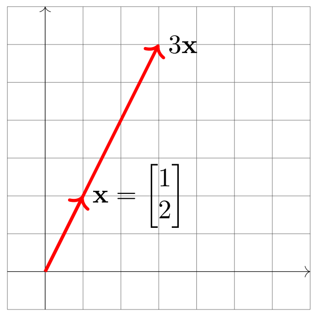
Given a square $n \times n$ matrix $\mathbf{A}$, we say that a nonzero vector, $\mathbf{x}$, is an eigenvector of $\mathbf{A}$ if there is a scalar λ such that $\mathbf{Ax} = \lambda\mathbf{x}$. The scalar λ is called the eigenvalue associated to the eigenvector $\mathbf{v}$.

Example 2.34 Eigenvector of a 2×2 Matrix

$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector of $\mathbf{A} = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$ corresponding to the eigenvector $\lambda = 3$.

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3\mathbf{x}$$

Geometrically, $\mathbf{x}$ multiplied by $\mathbf{A}$ stretches $\mathbf{x}$ by a factor of 3, as in



Computing Eigenvectors and Eigenvalues

If $\mathbf{A}$ is an $n \times n$ matrix,

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$$

In order for λ to be an eigenvector, $\mathbf{A} - \lambda\mathbf{I}$ must be singular. In other words, $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$. We can solve the charateristic equation for λ to get n solutions. We can use Gaussian elimination to find the null space of $\mathbf{A} - \lambda\mathbf{I}$. The vectors in the null space are eigenvectors of $\mathbf{A}$ with eigenvalue λ.

Example 2.35 Find the eigenvalues and eigenvectors of $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = 0$$

Thus, $\lambda_1 = 4, \lambda_2 = 2$

For $\lambda_1 = 4$, in the null space $\mathbf{A} - \lambda_1\mathbf{I} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$, the eigenvector $x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

For $\lambda_2 = 2$, in the null space $\mathbf{A} - \lambda_1\mathbf{I} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, the eigenvector $x_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Note that the coefficient 6 in the charateristic polymonial is the trace (sum of diagonal entries), and 8 is the determinant of $\mathbf{A}$. In general, the trace is the sum of the eigenvalues of a matrix, the product of the eigenvalues of any matrix equals to its determinant.

2.8 Step-by-step PCA for A Toy Example

Principal component analysis (PCA) is a linear dimensionality reduction technique for high dimensional data. The goal of PCA is to reduce the dimensionality of a high dimensional dataset in which there is a large number of variables while retaining as much as possible the variation in the original set of variables. The reduction of dimensionality is achievd by tranforming the original variables to a new set of principal components, which are uncorrelated small number of variables such that the first few components retain most of the variation present in the dataset. Algebraically, the principal components are linear combinations of the original variables- the first component has the largest variance and the second principal component has the second largest variance and is orthogonal to the first principal component. Geometrically, PCA represents a selection of a new coordinate system obtained by rotating the original axes to a set of new axes with a simpler structure. The first principal component represents the direction of the maximum variablity, and the second principal component represents the second maximum variability that is orthogonal to the first.

	veggies	rice	bread	milk
Alice	8	9	1	1
Bob	1	2	9	5
Cathy	7	8	3	4
Dylan	4	5	7	6
Emily	1	2	9	3

Table 2.1: Favorite Foods

In the table above, we have 5 data points (or called observations) and 4 dimensions (or called features). PCA is unsupervised learning so we do not need labels of any sort.

1. Standardize the data: Standardization rescales data to have a mean of zero and a standard deviation of 1. Here's how we compute each sample mean:

$$\overline{\mathbf{x}}_1 = \frac{8+1+7+4+1}{5} = 4.2$$

$$\overline{\mathbf{x}}_2 = \frac{9+2+8+5+2}{5} = 5.2$$

$$\overline{\mathbf{x}}_3 = \frac{1+9+3+7+9}{5} = 5.8$$

$$\overline{\mathbf{x}}_4 = \frac{1+5+4+6+3}{5} = 3.8$$

Thus, the average of the data is

$$\overline{\mathbf{x}} = \begin{bmatrix} 4.2 \\ 5.2 \\ 5.8 \\ 3.8 \end{bmatrix}$$

Here's how we compute each sample variance:

$$\hat{\sigma}_1^2 = \frac{(8-4.2)^2+(1-4.2)^2+(7-4.2)^2+(4-4.2)^2+(1-4.2)^2}{4} = 10.7$$

$$\hat{\sigma}_2^2 = \frac{(9-5.2)^2+(2-5.2)^2+(8-5.2)^2+(5-5.2)^2+(2-5.2)^2}{4} = 10.7$$

$$\hat{\sigma}_3^2 = \frac{(1-5.8)^2+(9-5.8)^2+(3-5.8)^2+(7-5.8)^2+(9-5.8)^2}{4} = 13.2$$

$$\hat{\sigma}_4^2 = \frac{(1-5.2)^2 + (5-5.2)^2 + (4-5.2)^2 + (6-5.2)^2 + (3-5.2)^2}{4} = 3.7$$

Thus, the standard deviation of data is

$$\sigma = \begin{bmatrix} 3.27 \\ 3.27 \\ 3.63 \\ 1.92 \end{bmatrix}$$

An example for computing the scaled data:

$$\frac{8-4.2}{3.27} = 1.162$$

We apply the formula for each feature and transform the original data to scaled data:

	veggies	rice	bread	milk
Alice	1.162	1.162	-1.321	-1.456
Bob	-0.978	-0.978	0.881	0.624
Cathy	0.856	0.856	-0.771	0.104
Dylan	-0.061	-0.061	0.330	1.144
Emily	-0.978	-0.978	0.881	-0.416

Table 2.2: Scaled Data

2. Compute covariance matrix of the scaled data: covariance matrix shows how variables relate to one another.

Here's how to compute covariance:

Let Z is the scaled variable data. For covariance of scaled variables:

$$\text{Cov}(Z_i, Z_j) = \frac{\text{Cov}(X_i, X_j)}{s_i s_j}$$

$$\text{Cov}(Z) = \frac{1}{n-1} Z^T Z$$

Since n=5:

$$\text{Cov}(Z) = \frac{1}{4} Z^T Z$$

We have the covariance matrix:

$$\text{Cov}(Z) = \begin{bmatrix} 1 & 1 & -0.9845 & -0.4688 \\ 1 & 1 & -0.9845 & -0.4688 \\ -0.9845 & -0.9845 & 1 & 0.6010 \\ -0.4688 & -0.4688 & 0.6010 & 1 \end{bmatrix}$$

3. Find the eigenvalues and eigenvectors of the covariance matrix. The eigenvectors give the direction of principal components, and the eigenvalues tells their importance.

Using the covariance matrix of the scaled data,

$$C_Z = \begin{bmatrix} 1 & 1 & -0.9845 & -0.4688 \\ 1 & 1 & -0.9845 & -0.4688 \\ -0.9845 & -0.9845 & 1 & 0.6010 \\ -0.4688 & -0.4688 & 0.6010 & 1 \end{bmatrix},$$

the eigenvectors are obtained by solving

$$C_Z \mathbf{v} = \lambda \mathbf{v}.$$

The eigenvalues, ordered from largest to smallest, are approximately

$$\lambda_1 = 3.3198, \qquad \lambda_2 = 0.6761, \qquad \lambda_3 = 0.0041, \qquad \lambda_4 \approx 0.$$

The corresponding eigenvectors are

$$\mathbf{v}_3 = \begin{bmatrix} 0.3762 \\ 0.3762 \\ 0.8336 \\ -0.1488 \end{bmatrix},$$

and

$$\mathbf{v}_4 = \begin{bmatrix} 0.7071 \\ -0.7071 \\ 0 \\ 0 \end{bmatrix}.$$

The variables are ordered as

$$[\text{veggies}, \text{rice}, \text{bread}, \text{milk}].$$

Therefore, the first principal component is

$$PC_1 = 0.5351 Z_{\text{veggies}} + 0.5351 Z_{\text{rice}} - 0.5469 Z_{\text{bread}} - 0.3580 Z_{\text{milk}},$$

and the second principal component is

$$PC_2 = 0.2686Z_{\text{veggies}} + 0.2686Z_{\text{rice}} - 0.0778Z_{\text{bread}} + 0.9218Z_{\text{milk}}.$$

Since the sum of the eigenvalues is approximately 4, the percentage of variance explained by each principal component is

$$\frac{\lambda_1}{4} \times 100 = \frac{3.3198}{4} \times 100 \approx 83.00\%,$$

and

$$\frac{\lambda_2}{4} \times 100 = \frac{0.6761}{4} \times 100 \approx 16.90\%.$$

Thus, the first two principal components explain approximately

$$83.00\% + 16.90\% = 99.90\%$$

of the total variance.

4. Project the scaled data from 4 dimensions into 2 dimensions.

Keep the first two eigenvectors:

$$\mathbf{v}_1 = \begin{bmatrix} 0.5351 \\ 0.5351 \\ -0.5469 \\ -0.3580 \end{bmatrix}$$

$$\mathbf{v}_2 = \begin{bmatrix} 0.2686 \\ 0.2686 \\ -0.0778 \\ 0.9218 \end{bmatrix}$$

From the projection matrix,

$$Y = ZW$$

where

$$W = \begin{bmatrix} 0.5351 & 0.2686 \\ 0.5351 & 0.2686 \\ -0.5469 & -0.0778 \\ -0.3580 & 0.9218 \end{bmatrix}.$$

Thus,

$$Y = \begin{bmatrix} 2.4870 & -0.6150 \\ -1.7520 & -0.0189 \\ 1.3004 & 0.6156 \\ -0.6555 & 0.9957 \\ -1.3798 & -0.9774 \end{bmatrix}.$$

For example, for Alice:

$$PC_{1,\text{Alice}} = (1.1617)(0.5351)+(1.1617)(0.5351)+(-1.3212)(-0.5469)+(-1.4557)(-0.3580)$$

$$PC_{2,\text{Alice}} = (1.1617)(0.2686)+(1.1617)(0.2686)+(-1.3212)(-0.0778)+(-1.4557)(-0.9218)$$

Thus,

PC1	PC2
2.4870	-0.6150
-1.7520	-0.0189
1.3004	0.6156
-0.6555	0.9957
-1.3798	-0.9774

These are the locations of the five people in the new two-dimensional PCA space.